

MYSQL VIEWS

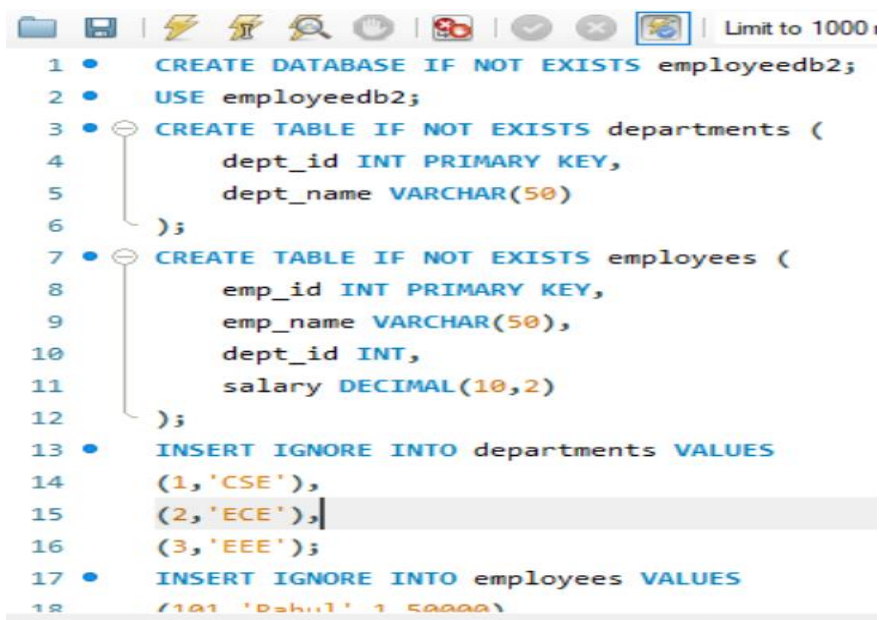
Practical Examples and Student Exercises using MySQL Workbench

1. What is a View?

A **View** is a **virtual table** created from one or more existing tables.

- It **does not store data** itself.
- It stores only the **SQL query**.

2. Setup: Employees and Departments Tables



```
1 • CREATE DATABASE IF NOT EXISTS employeedb2;
2 • USE employeedb2;
3 • CREATE TABLE IF NOT EXISTS departments (
4     dept_id INT PRIMARY KEY,
5     dept_name VARCHAR(50)
6 );
7 • CREATE TABLE IF NOT EXISTS employees (
8     emp_id INT PRIMARY KEY,
9     emp_name VARCHAR(50),
10    dept_id INT,
11    salary DECIMAL(10,2)
12 );
13 • INSERT IGNORE INTO departments VALUES
14     (1, 'CSE'),
15     (2, 'ECE'),
16     (3, 'EEE');
17 • INSERT IGNORE INTO employees VALUES
18     (101, 'Rahul', 1, 50000);
```

```
10    dept_id INT,
11    salary DECIMAL(10,2)
12 );
13 • INSERT IGNORE INTO departments VALUES
14     (1, 'CSE'),
15     (2, 'ECE'),
16     (3, 'EEE');
17 • INSERT IGNORE INTO employees VALUES
18     (101, 'Rahul', 1, 50000),
19     (102, 'Priya', 2, 65000),
20     (103, 'Anil', 1, 55000),
21     (104, 'Sneha', 3, 70000),
22     (105, 'Kiran', 2, 48000);
```

3. Verify the Tables

22 (105, 'Kiran', 2, 48000);

23

24 • SHOW TABLES;

25 • SELECT * FROM departments;

26 • SELECT * FROM employees;

27

28

Result Grid

Tables_in_employeedb2
departments
employees

24 • SHOW TABLES;

25 • SELECT * FROM departments;

26 • SELECT * FROM employees;

Result Grid

dept_id	dept_name
1	CSE
2	ECE
3	EEE
NULL	NULL

Result Grid

emp_id	emp_name	dept_id	salary
101	Rahul	1	50000.00
102	Priya	2	65000.00
103	Anil	1	55000.00
104	Sneha	3	70000.00
105	Kiran	2	48000.00

View 1 – Display All Employees

Question: Create a View named EmployeeView to display all columns from employees.

27

28 • DROP VIEW IF EXISTS EmployeeView;

29

30 • CREATE VIEW EmployeeView AS

31 SELECT *

32 FROM employees;

33

34 • SELECT * FROM EmployeeView;

Result Grid

emp_id	emp_name	dept_id	salary
101	Rahul	1	50000.00
102	Priya	2	65000.00
103	Anil	1	55000.00
104	Sneha	3	70000.00
105	Kiran	2	48000.00

EmployeeView 4 x

View 2 – Selected Columns

Question: Create a View showing employee ID, employee name and salary only.

The screenshot shows the SQL Developer interface. The SQL Editor contains the following code:

```
35  
36 DROP VIEW IF EXISTS EmployeeBasicView;  
37 CREATE VIEW EmployeeBasicView AS  
38 SELECT emp_id, emp_name, salary  
39 FROM employees;  
40 SELECT * FROM EmployeeBasicView;  
41  
42
```

Below the editor, the 'Result Grid' displays the output of the query. It shows a table with 5 rows and 3 columns: emp_id, emp_name, and salary.

emp_id	emp_name	salary
101	Rahul	50000.00
102	Priya	65000.00
103	Anil	55000.00
104	Sneha	70000.00
105	Kiran	48000.00

The bottom of the screenshot shows a tab labeled 'EmployeeBasicView 5'.

View 3 – High Salary Employees

Question: Create a View that displays employees earning more than 55000.

The screenshot shows the SQL Developer interface. The SQL Editor contains the following code:

```
42 DROP VIEW IF EXISTS HighSalaryEmployees;  
43 CREATE VIEW HighSalaryEmployees AS  
44 SELECT emp_id, emp_name, salary  
45 FROM employees  
46 WHERE salary > 55000;  
47 SELECT * FROM HighSalaryEmployees;  
48  
49
```

Below the editor, the 'Result Grid' displays the output of the query. It shows a table with 2 rows and 3 columns: emp_id, emp_name, and salary.

emp_id	emp_name	salary
102	Priya	65000.00
104	Sneha	70000.00

The bottom of the screenshot shows a tab labeled 'Setup'.

View 4 – Employee and Department using JOIN

Question: Create a View that displays employee name, department name and salary.

SQL Editor

```

50 • CREATE VIEW EmployeeDepartmentView AS
51 SELECT
52     e.emp_id,
53     e.emp_name,
54     d.dept_name,
55     e.salary
56 FROM employees e
57 JOIN departments d
58 ON e.dept_id = d.dept_id;
59 • SELECT * FROM EmployeeDepartmentView;

```

Result Grid

	emp_id	emp_name	dept_name	salary
▶	101	Rahul	CSE	50000.00
	102	Priya	ECE	65000.00
	103	Anil	CSE	55000.00
	104	Sneha	EEE	70000.00
	105	Kiran	ECE	48000.00

EmployeeDepartmentView 7

View 5 – Employees from a Particular Department

Question: Create a View that displays only CSE employees.

SQL Editor

```

61 • DROP VIEW IF EXISTS CSEEmployees;
62 • CREATE VIEW CSEEmployees AS
63 SELECT
64     e.emp_id,
65     e.emp_name,
66     d.dept_name,
67     e.salary
68 FROM employees e
69 JOIN departments d
70 ON e.dept_id = d.dept_id
71 WHERE d.dept_name = 'CSE';
72 • SELECT * FROM CSEEmployees;

```

Result Grid

	emp_id	emp_name	dept_name	salary
▶	101	Rahul	CSE	50000.00
	103	Anil	CSE	55000.00

Result Grid

	emp_id	emp_name	dept_name	salary
▶	101	Rahul	CSE	50000.00
	103	Anil	CSE	55000.00

View 6 – Department-wise Average Salary

Question: Create a View using GROUP BY and AVG() to display department-wise average salary.

The screenshot shows the SQL Developer interface. The SQL Editor contains the following code:

```
75 • CREATE VIEW DepartmentSalaryView AS
76 SELECT
77     d.dept_name,
78     COUNT(e.emp_id) AS EmployeeCount,
79     AVG(e.salary) AS AverageSalary
80 FROM employees e
81 JOIN departments d
82 ON e.dept_id = d.dept_id
83 GROUP BY d.dept_id, d.dept_name;
84 • SELECT * FROM DepartmentSalaryView;
```

Below the editor, the 'Result Grid' tab is active, displaying the following data:

dept_name	EmployeeCount	AverageSalary
CSE	2	52500.000000
ECE	2	56500.000000
EEE	1	70000.000000

View 7 – Departments with Average Salary Above 55000

Question: Create a View that displays departments whose average salary is greater than 55000.

The screenshot shows the SQL Developer interface. The SQL Editor contains the following code:

```
87 • CREATE VIEW HighAverageDepartments AS
88 SELECT
89     d.dept_name,
90     AVG(e.salary) AS AverageSalary
91 FROM employees e
92 JOIN departments d
93 ON e.dept_id = d.dept_id
94 GROUP BY d.dept_id, d.dept_name
95 HAVING AVG(e.salary) > 55000;
96 • SELECT * FROM HighAverageDepartments;
```

Below the editor, the 'Result Grid' tab is active, displaying the following data:

dept_name	AverageSalary
ECE	56500.000000
EEE	70000.000000

View 8 – Salary Range View

Question: Create a View that displays employees with salary between 50000 and 70000.

```

97
98 • DROP VIEW IF EXISTS SalaryRangeView;
99
100 • CREATE VIEW SalaryRangeView AS
101     SELECT emp_id, emp_name, salary
102     FROM employees
103     WHERE salary BETWEEN 50000 AND 70000;
104
105 • SELECT * FROM SalaryRangeView;
106

```

Result Grid | Filter Rows: | Export: | Wr

	emp_id	emp_name	salary
▶	101	Rahul	50000.00
	102	Priya	65000.00
	103	Anil	55000.00
	104	Sneha	70000.00

View 9 – View with Calculated Column

Question: Create a View that displays employee name, monthly salary and annual salary.

```

107 • DROP VIEW IF EXISTS AnnualSalaryView;
108 • CREATE VIEW AnnualSalaryView AS
109     SELECT
110         emp_id,
111         emp_name,
112         salary AS MonthlySalary,
113         salary * 12 AS AnnualSalary
114     FROM employees;
115 • SELECT * FROM AnnualSalaryView;
116

```

Result Grid | Filter Rows: | Export: | Wr

	emp_id	emp_name	MonthlySalary	AnnualSalary
▶	101	Rahul	50000.00	600000.00
	102	Priya	65000.00	780000.00
	103	Anil	55000.00	660000.00
	104	Sneha	70000.00	840000.00
	105	Kiran	48000.00	576000.00

AnnualSalaryView 12 x

View 10 – Update Data Through a Simple View

Question: Create a simple single-table View and update an employee's salary through the View.

```

116
117 • DROP VIEW IF EXISTS EmployeeSalaryView;
118 • CREATE VIEW EmployeeSalaryView AS
119     SELECT emp_id, emp_name, salary
120     FROM employees;
121 • UPDATE EmployeeSalaryView
122     SET salary = 60000
123     WHERE emp_id = 101;
124 • SELECT * FROM EmployeeSalaryView;
125 • SELECT * FROM employees WHERE emp_id = 101;

```

Result Grid

emp_id	emp_name	dept_id	salary
101	Rahul	1	60000.00
NULL	NULL	NULL	NULL

View 11 – CREATE OR REPLACE VIEW

Question: Modify an existing View so that it displays employee name and salary only.

```

127 • CREATE OR REPLACE VIEW EmployeeBasicView AS
128     SELECT emp_name, salary
129     FROM employees;
130
131 • SELECT * FROM EmployeeBasicView;
132
133
134

```

Result Grid

emp_name	salary
Rahul	60000.00
Priya	65000.00
Anil	55000.00
Sneha	70000.00
Kiran	48000.00

View 12 – View Definition

Question: Display the SQL statement used to create a View.


```

132
133 • SHOW CREATE VIEW EmployeeDepartmentView;
134

```

Result Grid				
	Filter Rows:		Export:	Wrap Cell Content:
View	Create View	character_set_client	collation_connection	
▶ employeedepartmentview	CREATE ALGORITHM=UNDEFINED DEFINER='r...	utf8mb4	utf8mb4_0900_ai_ci	

View 13 – List All Views

Question: Display all Views available in the current database.

```

134
135 • SHOW FULL TABLES
136 WHERE Table_type = 'VIEW';
137
138

```

Result Grid		
	Filter Rows:	Export:
Tables_in_employeedb2	Table_type	
▶ annualsalaryview	VIEW	
cseemployees	VIEW	
departmentsalaryview	VIEW	
employeebasicview	VIEW	
employeedepartmentview	VIEW	

Result 17 x

View 14 – Drop a View

Question: Remove an existing View from the database.

```

138 • DROP VIEW IF EXISTS SalaryRangeView;

```

Output

Action Output

#	Time	Action	Message
✓ 40	10:19:17	SELECT * FROM employees WHERE emp_id = 101 LIMIT 0, 1000	1 row(s) returned
✓ 41	10:22:41	CREATE OR REPLACE VIEW EmployeeBasicView AS SELECT emp_name, salary...	0 row(s) affected
✓ 42	10:22:41	SELECT * FROM EmployeeBasicView LIMIT 0, 1000	5 row(s) returned
✓ 43	10:23:27	SHOW CREATE VIEW EmployeeDepartmentView	1 row(s) returned
✓ 44	10:24:05	SHOW FULL TABLES WHERE Table_type = 'VIEW'	10 row(s) returned
✓ 45	10:24:38	DROP VIEW IF EXISTS SalaryRangeView	0 row(s) affected

6. Student Practice Tasks

1. Create a View showing only emp_id, emp_name and salary.

The screenshot shows the SQL Developer interface. The SQL Editor contains the following code:

```
35
36 • DROP VIEW IF EXISTS EmployeeBasicView;
37 • CREATE VIEW EmployeeBasicView AS
38   SELECT emp_id, emp_name, salary
39   FROM employees;
40 • SELECT * FROM EmployeeBasicView;
41
42
```

Below the editor, the 'Result Grid' tab is active, displaying the following data:

emp_id	emp_name	salary
101	Rahul	50000.00
102	Priya	65000.00
103	Anil	55000.00
104	Sneha	70000.00
105	Kiran	48000.00

The bottom of the window shows a tab labeled 'EmployeeBasicView 5'.

2. Create a View for employees earning more than 60000.

The screenshot shows the SQL Developer interface. The SQL Editor contains the following code:

```
139
140 • DROP VIEW IF EXISTS HighSalaryEmployee;
141 • CREATE VIEW HighSalaryEmployee AS
142   SELECT emp_id, emp_name, salary
143   FROM employees
144   WHERE salary > 60000;
145 • SELECT * FROM HighSalaryEmployee;
```

Below the editor, the 'Result Grid' tab is active, displaying the following data:

emp_id	emp_name	salary
102	Priya	65000.00
104	Sneha	70000.00

3. Create a View showing employee name, department name and salary using JOIN.

```

146
147 • CREATE VIEW employee_department_view AS
148 SELECT e.emp_name, d.dept_name, e.salary
149 FROM employees e
150 JOIN departments d
151 ON e.dept_id = d.dept_id;
152 • SELECT * FROM employee_department_view;

```

Result Grid | Filter Rows: | Export: | Wra

emp_name	dept_name	salary
Rahul	CSE	60000.00
Priya	ECE	65000.00
Anil	CSE	55000.00
Sneha	EEE	70000.00
Kiran	ECE	48000.00

employee_department_view 19 x

4. Create a View showing only ECE employees.

```

154 • DROP VIEW IF EXISTS ECEEmployees;
155 • CREATE VIEW ECEEmployees AS
156 SELECT
157     e.emp_id, e.emp_name,
158     d.dept_name, e.salary
159 FROM employees e
160 JOIN departments d
161 ON e.dept_id = d.dept_id
162 WHERE d.dept_name = 'ECE';
163 • SELECT * FROM ECEEmployees;

```

Result Grid | Filter Rows: | Export: | Limit to 100

emp_id	emp_name	dept_name	salary
102	Priya	ECE	65000.00
105	Kiran	ECE	48000.00

5. Create a View showing department-wise employee count.

bles

```

164
165 • CREATE VIEW dept_employee_count AS
166 SELECT d.dept_name, COUNT(e.emp_id) AS employee_count
167 FROM departments d
168 LEFT JOIN employees e
169 ON d.dept_id = e.dept_id
170 GROUP BY d.dept_name;
171 • SELECT * FROM dept_employee_count;

```

Result Grid

	dept_name	employee_count
▶	CSE	2
	ECE	2
	EEE	1

Export: Wrap Cell Content:

Setup

6. Create a View showing department-wise maximum salary.

les

```

172
173 • CREATE VIEW dept_max_salary AS
174 SELECT d.dept_name, MAX(e.salary) AS max_salary
175 FROM departments d
176 JOIN employees e
177 ON d.dept_id = e.dept_id
178 GROUP BY d.dept_name;
179 • SELECT * FROM dept_max_salary;

```

Result Grid

	dept_name	max_salary
▶	CSE	60000.00
	ECE	65000.00
	EEE	70000.00

Export: Wrap Cell Conte

Setup

7. Create a View showing departments whose average salary is greater than 50000.

```

180
181 • DROP VIEW IF EXISTS HighSalaryEmployeee;
182 • CREATE VIEW HighSalaryEmployeee AS
183     SELECT emp_id, emp_name, salary
184     FROM employees
185     WHERE salary > 50000;
186 • SELECT * FROM HighSalaryEmployeee;

```

Result Grid | Filter Rows: | Export: | Wrap Cell C

	emp_id	emp_name	salary
▶	101	Rahul	60000.00
	102	Priya	65000.00
	103	Anil	55000.00
	104	Sneha	70000.00

HighSalaryEmployeee 23 x

8. Create a View with employee name, salary and annual salary.

```

188 • DROP VIEW IF EXISTS AnnualSalaryViews;
189 • CREATE VIEW AnnualSalaryViews AS
190     SELECT
191         emp_id,
192         emp_name,
193         salary * 12 AS AnnualSalary
194     FROM employees;
195 • SELECT * FROM AnnualSalaryViews;

```

Result Grid | Filter Rows: | Export: | Wrap Cell C

	emp_id	emp_name	AnnualSalary
▶	101	Rahul	720000.00
	102	Priya	780000.00
	103	Anil	660000.00
	104	Sneha	840000.00
	105	Kiran	576000.00

AnnualSalaryViews 24 x

9. Update an employee's salary through a simple View and verify the base table.

197 • CREATE VIEW EmployeeSalary AS
 198 SELECT emp_id, emp_name, salary
 199 FROM employees;
 200 • UPDATE EmployeeSalary
 201 SET salary = 65000
 202 WHERE emp_id = 101;
 203 • SELECT * FROM EmployeeSalary
 204 WHERE emp_id = 101;
 205 • SELECT * FROM employees
 206 WHERE emp_id = 101;

Result Grid | Filter Rows: | Export:

emp_id	emp_name	salary
101	Rahul	65000.00

202 WHERE emp_id = 101;
 203 • SELECT * FROM EmployeeSalary
 204 WHERE emp_id = 101;
 205 • SELECT * FROM employees
 206 WHERE emp_id = 101;

Result Grid | Filter Rows: | Edit: | Export/

emp_id	emp_name	dept_id	salary
101	Rahul	1	65000.00
NULL	NULL	NULL	NULL

10. Use SHOW CREATE VIEW to display the definition of a View.

207
 208 • SHOW CREATE VIEW AnnualSalaryViews;

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

View	Create View	character_set_client	collation_connection
annualsalaryviews	CREATE ALGORITHM=UNDEFINED DEFINER='r...' utf8mb4	utf8mb4	utf8mb4_0900_ai_ci

11. List all Views in the database.

209

210 • `SHOW FULL TABLES WHERE TABLE_TYPE = 'VIEW';`

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	Tables_in_employeedb2	Table_type
▶	annualsalaryview	VIEW
	annualsalaryviews	VIEW
	csemployees	VIEW
	departmentsalaryview	VIEW
	dept_employee_count	VIEW

Result 28 x

12. Drop a View and verify that it no longer appears in the list.

211

212 • `DROP VIEW IF EXISTS AnnualSalaryViews;`

213 • `SHOW FULL TABLES WHERE TABLE_TYPE = 'VIEW';`

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	Tables_in_employeedb2	Table_type
▶	annualsalaryview	VIEW
	csemployees	VIEW
	departmentsalaryview	VIEW
	dept_employee_count	VIEW
	dept_max_salary	VIEW

Result 30 x

7. Viva / Discussion Questions

1. What is a View?

A View is a virtual table created from a SQL query. It displays selected data from one or more tables.

2. Why do we use Views?

Views are used to simplify complex queries, improve security, and show only required data.

3. Does a View normally store a separate copy of the data?

No. A View normally stores only the SQL query, not a separate copy of the data.

4. What is the difference between a View and a table?

A table stores actual data, while a View displays data from one or more tables.

5. Can a View be created using multiple tables?

Yes. A View can use multiple tables with JOINS.

6. Can every View be updated?

No. Simple Views can usually be updated, but Views using GROUP BY, aggregate functions, etc. may not be directly updatable.

7. Why are Views useful for security?

Views can hide sensitive columns and allow users to access only the data they need.

8. What is the difference between CREATE VIEW and CREATE OR REPLACE VIEW?

CREATE VIEW creates a new View.

CREATE OR REPLACE VIEW creates a new View or replaces an existing View.

9. How do you display the definition of a View in MySQL?

SHOW CREATE VIEW view_name;

It shows the SQL query used to create the View.

10. How do you remove a View?

DROP VIEW view_name;

It permanently removes the View, but does not delete the base table data.